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661 (continued): ON THE DOUBLE ϑ -FUNCTIONS.

venient form, we assume $\Omega\delta^2\Omega - (\delta\Omega)^2 = \Omega^2 M (\delta u)^2$, where M is a function of x . We thus obtain an equation $A\delta^2 A - (\delta A)^2 = \Omega^2 \mathfrak{A} (\delta u)^2$, where the value of $\mathfrak{A}$ depends upon that of M . The value of M has to be taken so as to simplify as much as may be the expression of $\mathfrak{A}$, but so that $\mathfrak{A}$ shall be a symmetrical function of the constants a, b, c, d : this last condition is assigned in order that the like simplification may present itself in the analogous relations $B\delta^2 B - (\delta B)^2 = \Omega^2 \mathfrak{B} (\delta u)^2$, &c. The proper expression of M is found to be

$$M = -2x^2 + x(a + b + c + d) + a^2 + b^2 + c^2 + d^2 - 2bc - 2ca - 2ab - 2ad - 2bd - 2cd,$$

viz. M having this value, the one other equation above referred to is

$$\Omega\delta^2\Omega - (\delta\Omega)^2 = \Omega^2 M (\delta u)^2;$$

and we then have a system of four equations such as

$$A\delta^2 A - (\delta A)^2 = \Omega^2 \mathfrak{A} (\delta u)^2,$$

where $\mathfrak{A}$ is a linear function of x , and where consequently $\Omega^2 \mathfrak{A}$ can be expressed as a linear function of any two of the four squares A^2, B^2, C^2, D^2 .

To establish the connexion with the Jacobian Θ and H functions, the differential relation between u, x may be taken to be

$$\delta u = \frac{\delta x}{\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

viz. we have here $d = \infty$, and to adapt the formulae to this value it is necessary to write $\frac{u}{\sqrt{d}}$ instead of u , and make other easy changes. The result is that Ω differs from D by a constant factor only, so that, instead of the five functions A, B, C, D, Ω , we have only the four functions A, B, C, D . The equation $\Omega\delta^2\Omega - (\delta\Omega)^2 = \Omega^2 \mathfrak{A} (\delta u)^2$ is replaced by an equation $D\delta^2 D - (\delta D)^2 = D^2 \mathfrak{D} (\delta u)^2$, or say $= \delta^2(\log D) D^2 (\delta u)^2$, which gives a result of the form

$$D = \Theta e^{\alpha u^2},$$

showing that D differs from Jacobi's $\Theta(u)$ only by an exponential factor of the form $\Theta e^{\alpha u^2}$. And it then further appears that A, B, C differ only by factors of the like form from the three numerator functions $H(u), H(u + K), \Theta(u + K)$, so that, neglecting constant factors, the functions

$$\frac{A}{D}, \quad \frac{B}{D}, \quad \frac{C}{D}$$

are equal to $\frac{H(u)}{\Theta(u)}, \frac{H(u + K)}{\Theta(u)}, \frac{\Theta(u + K)}{\Theta(u)}$, that is, to the elliptic functions $\text{sn } u, \text{cn } u, \text{dn } u$.

662.

ON THE DOUBLE ϑ -FUNCTIONS IN CONNEXION WITH A 16-NODAL QUARTIC SURFACE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXIII. (1877), pp. 210–219.]

I have before me Göpel's memoir, "Theoriae transcendentium Abelianarum primi ordinis ad-umbratio levis," Crelle's Journal, t. XXXV. (1847), pp. 277–312. Writing P_1, P_2, P_3, P_4 , etc., in place of his P', P'', P''' , etc., also $\alpha, \beta, \gamma, \delta, X', Y', Z', W'$, in place of his t, u, v, w, T, U, V, W , the system of 16 equations (given p. 287) is

$$\begin{aligned}
 (1) \quad P^2 &= (\alpha, -\beta, -\gamma, \delta)(X', Y', Z', W'), \\
 (4) \quad P_1^2 &= (\alpha, \beta, -\gamma, -\delta)(X', Y', Z', W'), \\
 (9) \quad P_2^2 &= (\alpha, -\beta, \gamma, -\delta)(X', Y', Z', W'), \\
 (12) \quad P_3^2 &= (\alpha, \beta, \gamma, \delta)(X', Y', Z', W'); \\
 (3) \quad Q^2 &= (\beta, -\alpha, -\delta, \gamma)(X', Y', Z', W'), \\
 (2) \quad Q_1^2 &= (\beta, \alpha, -\delta, -\gamma)(X', Y', Z', W'), \\
 (11) \quad Q_2^2 &= (\beta, -\alpha, \delta, -\gamma)(X', Y', Z', W'), \\
 (10) \quad Q_3^2 &= (\beta, \alpha, \delta, \gamma)(X', Y', Z', W'); \\
 (13) \quad R^2 &= (\gamma, -\delta, -\alpha, \beta)(X', Y', Z', W'), \\
 (16) \quad R_1^2 &= (\gamma, \delta, -\alpha, -\beta)(X', Y', Z', W'), \\
 (5) \quad R_2^2 &= (\gamma, -\delta, \alpha, -\beta)(X', Y', Z', W'), \\
 (8) \quad R_3^2 &= (\gamma, \delta, \alpha, \beta)(X', Y', Z', W'); \\
 (15) \quad S^2 &= (\delta, -\gamma, -\beta, \alpha)(X', Y', Z', W'), \\
 (14) \quad S_1^2 &= (\delta, \gamma, -\beta, -\alpha)(X', Y', Z', W'), \\
 (7) \quad S_2^2 &= (\delta, -\gamma, \beta, -\alpha)(X', Y', Z', W'), \\
 (6) \quad S_3^2 &= (\delta, \gamma, \beta, \alpha)(X', Y', Z', W');
 \end{aligned}$$

viz. we have $P^2 = \alpha X' - \beta Y' - \gamma Z' + \delta W'$, etc. The reason for the apparently arbitrary manner in which I have numbered these equations, will appear further on. I recall that the sixteen double ϑ -functions, that is, ϑ -functions of two arguments u, u' , are¹

$$\begin{array}{cccc} P, & P_1, & P_2, & P_3, \\ iQ, & Q_1, & iQ_2, & Q_3, \\ iR, & iR_1, & R_2, & R_3, \\ S, & iS_1, & iS_2, & S_3, \end{array}$$

the factor $i, = \sqrt{-1}$, being introduced in regard to the six functions which are odd functions of the arguments. But disregarding the i sign, I speak of $P^2, P_1^2, \dots, Q^2$, etc., as the squared functions, or simply as the squares; $\alpha, \beta, \gamma, \delta$ are constants, depending of course on the parameters of the ϑ -functions; X', Y', Z', W' , which are however to be eliminated, are themselves ϑ -functions to a different set of parameters: the 16 equations express that the squared functions P^2, P_1^2 , etc., are linear functions of X', Y', Z', W' , and they consequently serve to obtain linear relations between the squared functions: viz. by means of them, Göpel expresses the remaining 12 squares, each in terms of the selected four squares $P_1^2, P_3^2, S_1^2, S_3^2$, which are linearly independent: that is, he obtains linear relations between five squares, and he seems to have assumed that there were not any linear relations between fewer than five squares.

It appears however by Rosenhain's "Mémoire sur les fonctions de deux variables et à quatre périodes etc.", Mém. Sav. Etrangers, t. XI. (1851), pp. 364–468, that there are, in fact, linear relations between four squares, viz. that there exist sixes of squares such that, selecting at pleasure any three out of the six, each of the remaining three squares can be expressed as a linear function of these three squares. Knowing this result, it is easy to verify it by means of the sixteen equations, and moreover to show that there are in all 16 such sixes: these are shown by the following scheme which I copy from Kummer's memoir "Ueber die algebraischen Strahlensysteme u. s. w." Berlin. Abh. (1866), p. 66: viz. the scheme is

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
9	10	11	12	13	14	15	16	1	2	3	4	5	6	7	8
13	14	15	16	9	10	11	12	5	6	7	8	1	2	3	4
8	7	6	5	4	3	2	1	16	15	14	13	12	11	10	9
7	8	5	6	3	4	1	2	15	16	13	14	11	12	9	10
6	5	8	7	2	1	4	3	14	13	16	15	10	9	12	11.

¹The same functions in Rosenhain's notation are

00, 02, 20, 22,
01, 03, 21, 23,
10, 12, 30, 32,
11, 13, 31, 33;

viz. the figures here written down are the suffixes of his ϑ -functions, $00 = \vartheta_{0,0}(v, w)$, etc.

In fact, to show that any four of the squares, for instance 1, 9, 13, 8, that is, P^2, P_2^2, R^2, R_3^2 , are linearly connected, it is only necessary to show that the determinant of coefficients

$$\begin{vmatrix} \alpha, & -\beta, & -\gamma, & \delta \\ \alpha, & -\beta, & \gamma, & -\delta \\ \gamma, & -\delta, & -\alpha, & \beta \\ \gamma, & \delta, & \alpha, & \beta \end{vmatrix}$$

is $= 0$, or what is the same thing, that there exists a linear function of the new variables (X, Y, Z, W) , which will become $= 0$ on putting for these variables the values in any line of this determinant: we have such a function, viz. this is

$$\beta X + \alpha Y - \delta Z - \gamma W,$$

or say

$$[1] (\beta, \alpha, -\delta, -\gamma)(X, Y, Z, W).$$

This function also vanishes if for (X, Y, Z, W) we substitute the values

$$\delta, \quad -\gamma, \quad -\beta, \quad -\alpha,$$

$$\delta, \quad \gamma, \quad \beta, \quad \alpha,$$

which belong to 7, 6, that is, S_2^2 and S_3^2 respectively. It thus appears that 1, 9, 13, 8, 7, 6, that is, $P^2, P_2^2, R^2, R_3^2, S_2^2, S_3^2$, are a set of six squares having the property in question. I remark that the process of forming the linear functions is a very simple one; we write down six lines, and thence directly obtain the result, thus

$$\begin{array}{rcll} 1 & \alpha, & -\beta, & -\gamma, & \delta \\ 9 & \alpha, & -\beta, & \gamma, & -\delta \\ 13 & \gamma, & -\delta, & -\alpha, & \beta \\ 8 & \gamma, & \delta, & \alpha, & \beta \\ 7 & \delta, & -\gamma, & \beta, & -\alpha \\ 6 & \delta, & \gamma, & \beta, & \alpha \\ & \beta, & \alpha, & -\delta, & -\gamma : \end{array}$$

viz. $\beta, \alpha, \delta, \gamma$ are the letters not previously occurring in the four columns respectively: the first letter β is taken to have the sign $+$, and then the remaining signs are determined by the condition that, combining the last line with any line above it (e.g. with the line next above it $\beta\delta + \alpha\gamma - \delta\beta - \gamma\alpha$), the sum must be zero.

We find in this way, as the conditions for the existence of the 16 sixes respectively,

$$\begin{array}{l} [1] (\beta, \alpha, -\delta, -\gamma)(X, Y, Z, W) = 0, \\ [2] (\alpha, -\beta, -\gamma, \delta)(X, Y, Z, W) = 0, \\ [3] (\alpha, \beta, -\gamma, -\delta)(X, Y, Z, W) = 0, \\ [4] (\beta, -\alpha, -\delta, \gamma)(X, Y, Z, W) = 0, \end{array}$$

- [5] $(\delta, \gamma, \beta, \alpha)(X, Y, Z, W) = 0,$
- [6] $(\gamma, -\delta, \alpha, -\beta)(X, Y, Z, W) = 0,$
- [7] $(\gamma, \delta, \alpha, \beta)(X, Y, Z, W) = 0,$
- [8] $(\delta, -\gamma, \beta, -\alpha)(X, Y, Z, W) = 0,$
- [9] $(\beta, \alpha, \delta, \gamma)(X, Y, Z, W) = 0,$
- [10] $(\alpha, -\beta, \gamma, -\delta)(X, Y, Z, W) = 0,$
- [11] $(\alpha, \beta, \gamma, \delta)(X, Y, Z, W) = 0,$
- [12] $(\beta, -\alpha, \delta, -\gamma)(X, Y, Z, W) = 0,$
- [13] $(\delta, -\gamma, -\beta, \alpha)(X, Y, Z, W) = 0,$
- [14] $(\gamma, -\delta, -\alpha, \beta)(X, Y, Z, W) = 0,$
- [15] $(\gamma, \delta, -\alpha, -\beta)(X, Y, Z, W) = 0,$
- [16] $(\delta, \gamma, -\beta, -\alpha)(X, Y, Z, W) = 0.$

I repeat in a new order the sets of coefficients which belong to the several squares, viz. these are

- (1) $P^2 = (\alpha, -\beta, -\gamma, \delta),$
- (2) $Q_1^2 = (\beta, \alpha, -\delta, -\gamma),$
- (3) $Q^2 = (\beta, -\alpha, -\delta, \gamma),$
- (4) $P_1^2 = (\alpha, \beta, -\gamma, -\delta),$
- (5) $R_2^2 = (\gamma, -\delta, \alpha, -\beta),$
- (6) $S_3^2 = (\delta, \gamma, \beta, \alpha),$
- (7) $S_2^2 = (\delta, -\gamma, \beta, -\alpha),$
- (8) $R_3^2 = (\gamma, \delta, \alpha, \beta),$
- (9) $P_2^2 = (\alpha, -\beta, \gamma, -\delta),$
- (10) $Q_3^2 = (\beta, \alpha, \delta, \gamma),$
- (11) $Q_2^2 = (\beta, -\alpha, \delta, -\gamma),$
- (12) $P_3^2 = (\alpha, \beta, \gamma, \delta),$
- (13) $R^2 = (\gamma, -\delta, -\alpha, \beta),$
- (14) $S_1^2 = (\delta, \gamma, -\beta, -\alpha),$
- (15) $S^2 = (\delta, -\gamma, -\beta, \alpha),$
- (16) $R_1^2 = (\gamma, \delta, -\alpha, -\beta).$

And I remark that, if we connect these with the multipliers $(Y, -X, W, -Z)$, we obtain, except that there is sometimes a reversal of all the signs, the same linear functions of (X, Y, Z, W) as are written down under the same numbers in square brackets above: thus (1) gives

$$(\alpha, -\beta, -\gamma, \delta)(Y, -X, W, -Z), \text{ which is } (\beta, \alpha, -\delta, -\gamma)(X, Y, Z, W), = [1];$$

and so (2) gives

$$(\beta, \alpha, -\delta, -\gamma)(Y, -X, W, -Z), \text{ which is } (-\alpha, \beta, \gamma, -\delta)(X, Y, Z, W),$$

or, reversing the signs,

$$(\alpha, -\beta, -\gamma, \delta)(X, Y, Z, W), = [2].$$

Comparing with the geometrical theory in Kummer's Memoir, it appears that the several systems of values (1), (2), . . . , (16) are the coordinates of the nodes of a 16-nodal quartic surface, which nodes lie by sixes in the singular tangent planes, in the manner expressed by the foregoing scheme, wherein each top number may refer to a singular tangent plane, and then the numbers below it show the nodes in this plane: or else the top number may refer to a node, and then the numbers below it show the singular planes through this node.

And, from what precedes, we have the general result: the 16 squared double ϑ -functions correspond (one to one) to the nodes of a 16-nodal quartic surface, in such wise that linearly connected squared functions correspond to nodes in the same singular tangent plane.

The question arises, to find the equation of the 16-nodal quartic surface, having the foregoing nodes and singular tangent planes. Starting from one of the irrational forms, say

$$\sqrt{A} [1][5] + \sqrt{B} [2][6] + \sqrt{C} [3][7] = 0,$$

the coefficients A, B, C are readily determined; and the result written at full length is

$$\begin{aligned} & \sqrt{2}(\alpha\beta - \gamma\delta)(\alpha\delta + \beta\gamma)(\beta X + \alpha Y - \delta Z - \gamma W)(\delta X + \gamma Y + \beta Z + \alpha W) \\ & + \sqrt{(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(\alpha\gamma - \beta\delta)}(\alpha X - \beta Y - \gamma Z + \delta W)(\gamma X - \delta Y - \alpha Z + \beta W) \\ & + \sqrt{(\alpha^2 - \beta^2 - \gamma^2 + \delta^2)(\alpha\gamma + \beta\delta)}(\alpha X + \beta Y - \gamma Z - \delta W)(\gamma X + \delta Y + \alpha Z + \beta W) = 0. \end{aligned}$$

It is a somewhat long, but nevertheless interesting, piece of algebraical work to rationalise the foregoing equation: the result is

$$\begin{aligned} & (\gamma^2\delta^2 - \alpha^2\beta^2)(\beta^2\gamma^2 - \alpha^2\delta^2)(\alpha^2\beta^2 - \gamma^2\delta^2)(X^4 + Y^4 + Z^4 + W^4) \\ & + (\gamma^2\delta^2 - \alpha^2\beta^2)(\alpha^2\beta^2 - \gamma^2\delta^2)(\alpha^2 + \delta^2 - \beta^2 - \gamma^2)(Y^2Z^2 + X^2W^2) \\ & + (\alpha^2\beta^2 - \gamma^2\delta^2)(\beta^2\gamma^2 - \alpha^2\delta^2)(\beta^2 + \delta^2 - \gamma^2 - \alpha^2)(Z^2X^2 + Y^2W^2) \\ & + (\beta^2\gamma^2 - \alpha^2\delta^2)(\gamma^2\delta^2 - \alpha^2\beta^2)(\gamma^2 + \delta^2 - \alpha^2 - \beta^2)(X^2Y^2 + Z^2W^2) \\ & + 2\alpha\beta\gamma\delta(\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(\alpha^2 - \beta^2 - \gamma^2 + \delta^2)XYZW = 0; \end{aligned}$$

or, if we write for shortness

$$\begin{aligned} L &= \beta^2\gamma^2 - \alpha^2\delta^2, & F &= \alpha^2 + \beta^2 - \gamma^2 - \delta^2, \\ M &= \gamma^2\alpha^2 - \beta^2\delta^2, & G &= \alpha^2 - \beta^2 + \gamma^2 - \delta^2, \\ N &= \alpha^2\beta^2 - \gamma^2\delta^2, & H &= \alpha^2 - \beta^2 - \gamma^2 + \delta^2, \\ & & \Delta &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2, \end{aligned}$$

then the result is

$$\begin{aligned} & LMN(X^4 + Y^4 + Z^4 + W^4) \\ & + MN(F\Delta + 2L)(Y^2Z^2 + X^2W^2) \\ & + NL(G\Delta + 2M)(Z^2X^2 + Y^2W^2) \\ & + LM(H\Delta + 2N)(X^2Y^2 + Z^2W^2) \\ & - 2\alpha\beta\gamma\delta FGH\Delta XYZW = 0. \end{aligned}$$

It may be easily verified that any one of the sixteen points, for instance $(\alpha, \beta, \gamma, \delta)$, is a node of the surface. Thus to show that the derived function in respect to X , vanishes for $X, Y, Z, W = \alpha, \beta, \gamma, \delta$; the derived function here divides by 2α , and omitting this factor, the equation to be verified is

$$LMN \cdot 2\alpha^2 + MN(F\Delta + 2L)\delta^2 + NL(G\Delta + 2M)\gamma^2 + LM(H\Delta + 2N)\beta^2 - \beta^2\gamma^2\delta^2FGH\Delta = 0,$$

viz. the whole coefficient of LMN is $2(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)$, $= 2\Delta$; hence throwing out the factor Δ , the equation becomes

$$2LMN + MNF\delta^2 + NLG\gamma^2 + LMH\beta^2 - \beta^2\gamma^2\delta^2FGH = 0.$$

Writing this in the form

$$L(2MN + NG\gamma^2 + MH\beta^2) = \beta^2\delta^2(\gamma^2\delta^2FGH - MNF),$$

we find without difficulty $GH\gamma^2\delta^2 - MN = -L(\delta^2 - \gamma^2)^2$; hence throwing out the factor L , the equation becomes

$$G(2M + G\gamma^2) + MH\beta^2 + F\delta^2(\delta^2 - \gamma^2)^2 = 0;$$

we find

$$\begin{aligned} MH\beta^2 + F\delta^2(\delta^2 - \gamma^2)^2 &= (\alpha^2\beta^2 - \gamma^2\delta^2)(2\delta^2\beta^2 - \gamma^2(\alpha^2 + \beta^2 + \delta^2) + \gamma^4) \\ &= -N(2\beta^2\delta^2 - \gamma^2(\alpha^2 + \beta^2 + \delta^2) + \gamma^4), \end{aligned}$$

or throwing out the factor N , the equation becomes

$$2M + G\gamma^2 - 2\beta^2\delta^2 + \gamma^2(\alpha^2 + \beta^2 + \delta^2) - \gamma^4 = 0,$$

which is at once verified: and similarly it can be shown that the other derived functions vanish, and the point $(\alpha, \beta, \gamma, \delta)$ is thus a node.

The surface seems to be the general 16-nodal surface, viz. replacing X, Y, Z, W by any linear functions of four coordinates, we have thus $4 \cdot 4 - 1 = 15$ constants, and the equation contains besides the three ratios $\alpha : \beta : \gamma : \delta$, that is, in all 18 constants: the general quartic surface has 34 constants, and therefore the general 16-nodal surface $34 - 16 = 18$ constants: but the conclusion requires further examination.

Göpel and Rosenhain each connect the theory with that of the ultra-elliptic functions involving the radical $\sqrt{X}$, $= \sqrt{x \cdot 1 - x \cdot 1 - lx \cdot 1 - mx \cdot 1 - nx}$; viz. it appears by their formulae (more completely by those of Rosenhain) that the ratios of the 16 squares can be expressed rationally in terms of the two variables x, x' , and the radicals

$\sqrt{X}, \sqrt{X'}$, X' being the same function of x' that X is of x . We may instead of the preceding form take X to be the general quintic function, or what is better take it to be the sextic function $\overline{a-x} \cdot \overline{b-x} \cdot \overline{c-x} \cdot \overline{d-x} \cdot \overline{e-x} \cdot \overline{f-x}$; and we thus obtain a remarkable algebraical theorem: viz. I say that the 16 squares, each divided by a proper constant factor, are proportional to six functions of the form

$$\overline{a-x} \cdot \overline{a-x'},$$

and ten functions of the form

$$\frac{1}{(x-x')^2} \left\{ \sqrt{\overline{a-x} \cdot \overline{b-x} \cdot \overline{c-x} \cdot \overline{d-x'} \cdot \overline{e-x'} \cdot \overline{f-x'}} - \sqrt{\overline{a-x'} \cdot \overline{b-x'} \cdot \overline{c-x'} \cdot \overline{d-x} \cdot \overline{e-x} \cdot \overline{f-x}} \right\}^2,$$

and consequently that these 16 algebraical functions of x, x' are linearly connected in the manner of the 16 squares; viz. there exist 16 sixes such that, in each six, the remaining three functions can be linearly expressed in terms of any three of them.

To further develop the theory, I remark that the six functions may be represented by A, B, C, D, E, F respectively: any one of the ten functions would be properly represented by $\frac{ABC}{DEF}$, but isolating one letter F , and writing DEF to denote this function, may be represented simply as DE ; and the ten functions thus are $AB, AC, AD, AE, BC, BD, BE, CD, CE, DE$.

Writing for shortness a, b, c, d, e, f , to denote $a-x, b-x$, etc., and similarly a', b', c', d', e', f' , to denote $a-x', b-x'$, etc., we thus have

$$\begin{aligned} (13) \quad A &= aa', \quad (= E), \\ (9) \quad B &= bb', \\ (7) \quad C &= cc', \quad (= F), \\ (8) \quad D &= dd', \\ (6) \quad E &= ee', \\ (1) \quad F &= ff', \\ (3) \quad DE &= \frac{1}{(x-x')^2} \{ \sqrt{abcd'e'f'} - \sqrt{a'b'c'def} \}^2, \quad (= D), \\ (4) \quad CE &= \frac{1}{(x-x')^2} \{ \sqrt{abdc'e'f'} - \sqrt{a'b'd'cef} \}^2, \quad (= E), \\ (2) \quad CD &= \frac{1}{(x-x')^2} \{ \sqrt{abec'd'f'} - \sqrt{a'b'e'cdf} \}^2, \\ (14) \quad BE &= \frac{1}{(x-x')^2} \{ \sqrt{acdb'e'f'} - \sqrt{a'c'd'bef} \}^2, \quad (= B), \\ (16) \quad BD &= \frac{1}{(x-x')^2} \{ \sqrt{aceb'd'f'} - \sqrt{a'c'e'bdf} \}^2, \\ (15) \quad BC &= \frac{1}{(x-x')^2} \{ \sqrt{adeb'c'f'} - \sqrt{a'd'e'bcf} \}^2, \\ (10) \quad AE &= \frac{1}{(x-x')^2} \{ \sqrt{bcda'e'f'} - \sqrt{b'c'd'ae'f} \}^2, \quad (= I), \end{aligned}$$

$$\begin{aligned}
 (12) \quad AD &= \frac{1}{(x-x')^2} \{ \sqrt{bcea'd'f'} - \sqrt{b'e'ad'f} \}^2, \\
 (11) \quad AC &= \frac{1}{(x-x')^2} \{ \sqrt{bdea'c'f'} - \sqrt{b'd'e'acf} \}^2, \\
 (5) \quad AB &= \frac{1}{(x-x')^2} \{ \sqrt{cdea'b'f'} - \sqrt{c'd'e'abf} \}^2,
 \end{aligned}$$

where the numbers are in accordance with the foregoing scheme; viz. the scheme becomes

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
<i>F</i>	<i>CD</i>	<i>DE</i>	<i>CE</i>	<i>AB</i>	<i>E</i>	<i>C</i>	<i>D</i>	<i>B</i>	<i>AE</i>	<i>AC</i>	<i>AD</i>	<i>A</i>	<i>BE</i>	<i>BC</i>	<i>BD</i>
<i>B</i>	<i>AE</i>	<i>AC</i>	<i>AD</i>	<i>A</i>	<i>BE</i>	<i>BC</i>	<i>BD</i>	<i>F</i>	<i>CD</i>	<i>DE</i>	<i>CE</i>	<i>AB</i>	<i>E</i>	<i>C</i>	<i>D</i>
<i>C</i>	<i>D</i>	<i>A</i>	<i>BE</i>	<i>BC</i>	<i>BD</i>	<i>B</i>	<i>AE</i>	<i>AC</i>	<i>AD</i>	<i>AB</i>	<i>E</i>	<i>C</i>	<i>D</i>	<i>F</i>	<i>CD</i>
	<i>E</i>	<i>AB</i>	<i>CE</i>	<i>DE</i>	<i>CD</i>	<i>F</i>	<i>BD</i>	<i>BC</i>	<i>BE</i>	<i>A</i>	<i>AD</i>	<i>AC</i>	<i>AE</i>	<i>B</i>	<i>DE</i>
<i>D</i>	<i>AB</i>	<i>E</i>	<i>C</i>	<i>DE</i>	<i>CE</i>	<i>F</i>	<i>CD</i>	<i>BC</i>	<i>BD</i>	<i>A</i>	<i>BE</i>	<i>AC</i>	<i>AD</i>	<i>B</i>	<i>AE</i>
<i>E</i>	<i>AB</i>	<i>D</i>	<i>C</i>	<i>CD</i>	<i>F</i>	<i>CE</i>	<i>DE</i>	<i>DE</i>	<i>A</i>	<i>BD</i>	<i>BC</i>	<i>AE</i>	<i>B</i>	<i>AD</i>	<i>AC</i>

There is of course the six A, B, C, D, E, F ; for each of these is a linear function of $1, x + x', xx'$, and there is thus a linear relation between any four of them. It would at first sight appear that the remaining sixes were of two different forms, A, B, AB, CE, CD, DE , and F, A, AB, AC, AD, AE ; but these are really identical, for taking any two letters E, F , the six is E, F, AE, BE, CE, DE , or, as this might be written, E, F, AEF, BEF, CEF, DEF , where AEF means $BCD \cdot AEF$, etc.; and we thus obtain each of the remaining fifteen sixes. The six just referred to, viz. E, F, AE, BE, CE, DE , or changing the notation say E, F, A, B, C, D as indicated in the table, thus represents any one of the sixes other than the rational six A, B, C, D, E, F : and there is no difficulty in actually finding each of the fifteen relations between four functions of the six in question, E, F, A, B, C, D . It is to be observed that every such function as A contains the same irrational part

$$\frac{2}{(x-x')^2} \sqrt{abcdef a'b'c'd'e'f'},$$

and that the linear relations involve therefore only the differences $A - B, A - C$, etc., which are rational. Proceeding to calculate these differences, we have for instance

$$C - D = \frac{1}{(x-x')^2} (cefa'b'd' + c'e'fabd - defa'b'c' - d'e'f'abc) = \frac{1}{(x-x')^2} (cd' - c'd)(efa'b' - e'f'ab);$$

or, substituting for a, a' , etc. their values $a - x, a - x'$, etc., we have

$$cd' - c'd = (x - x')(c - d),$$

$$efa'b' - e'f'ab = -(x - x') \begin{vmatrix} 1, & x + x', & xx' \\ 1, & a + b, & ab \\ 1, & e + f, & ef \end{vmatrix} = (x - x')[xx'abef],$$

or say for shortness.

We have therefore

$$C - D = (c - d)[xx'abef];$$

and in like manner we obtain the equations

$$\begin{aligned} A - B &= (a - b)[xx'cdef], & C - D &= (c - d)[xx'abef], \\ A - C &= (a - c)[xx'bdef], & B - D &= (b - d)[xx'caef], \\ A - D &= (a - d)[xx'bcef], & B - C &= (b - c)[xx'adef]. \end{aligned}$$

It is now easy to form the system of formulae

<i>E</i>	<i>F</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
$ae \cdot af \cdot bcd$	$-be \cdot bf \cdot cda$	$+ce \cdot cf \cdot dab$	$-de \cdot df \cdot abc$		$= 0,$
$ad \cdot bf \cdot cf$	$-ad \cdot be \cdot ce$	$+ef$	$-ef$		$= 0,$
$bd \cdot cf \cdot af$	$-bd \cdot ce \cdot ae$	$+ef$	$-ef$		$= 0,$
$cd \cdot af \cdot bf$	$-cd \cdot ae \cdot be$	$+ef$	$-ef$		$= 0,$
$bc \cdot af \cdot df$	$-bc \cdot ae \cdot de$	$+ef$	$-ef$		$= 0,$
$ca \cdot bf \cdot df$	$-ca \cdot be \cdot de$	$+ef$	$-ef$		$= 0,$
$ab \cdot cf \cdot df$	$-ab \cdot ce \cdot de$	$+ef$	$-ef$		$= 0,$
$-af \cdot bcd$		$+be \cdot cd$	$+ce \cdot db$	$+de \cdot bc$	$= 0,$
$-bf \cdot cda$		$+ae \cdot cd$	$+ce \cdot da$	$+de \cdot ac$	$= 0,$
$-cf \cdot dab$	$+ae \cdot bd$	$+be \cdot ca$	$+ce \cdot ab$		$= 0,$
$-df \cdot abc$	$+ae \cdot bc$			$+de \cdot ab$	$= 0,$
$-ae \cdot bcd$		$+bf \cdot cd$	$+cf \cdot db$	$+df \cdot bc$	$= 0,$
$-be \cdot cda$		$+af \cdot cd$	$+cf \cdot da$	$+df \cdot ac$	$= 0,$
$-ce \cdot dab$	$+af \cdot bd$	$+bf \cdot ca$	$+cf \cdot ab$		$= 0,$
$-de \cdot abc$	$+af \cdot bc$			$+df \cdot ab$	$= 0,$

where for shortness ab, ac , etc., are written to denote $a - b, a - c$, etc.; also abc , etc., to denote $(b - c)(c - a)(a - b)$, etc.: the equations contain all of them only the differences of A, B, C, D ; thus the first equation is equivalent to

$$ae \cdot af \cdot bcd (A - D) - be \cdot bf \cdot cde (B - D) + ce \cdot cf \cdot dab (C - D) = 0,$$

and so in other cases.

Cambridge, 14 March, 1877.

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FURTHER INVESTIGATIONS ON THE DOUBLE ϑ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXIII. (1877), pp. 220–233.]

I consider six letters

$$a, b, c, d, e, f;$$

a duad ab not containing f may be completed into the triad abf , and then into the double triad abf, cde ; there are in all ten double triads, represented by the duads

$$ab, ac, ad, ae, bc, bd, be, cd, ce, de,$$

and the whole number of letters and of double triads is = 16.

Taking x, x' as variables, I form sixteen functions; viz. these are

$$[a] = \overline{a-x} \cdot \overline{a-x'},$$

$$[ab] = \frac{1}{(x-x')^2} \left\{ \sqrt{\frac{\overline{a-x} \cdot \overline{b-x} \cdot \overline{f-x}}{\overline{c-x'} \cdot \overline{d-x'} \cdot \overline{e-x'}}} \pm \sqrt{\frac{\overline{a-x'} \cdot \overline{b-x'} \cdot \overline{f-x'}}{\overline{c-x} \cdot \overline{d-x} \cdot \overline{e-x}}} \right\}^2,$$

where the function under each radical sign is the product of six factors, the arrangement in two lines being for convenience only: the sign $\pm$ has the same value in all the functions, and it will be observed that the irrational part is

$$\pm \frac{2}{(x-x')^2} \sqrt{\frac{\overline{a-x} \cdot \overline{b-x} \cdot \overline{c-x} \cdot \overline{d-x} \cdot \overline{e-x} \cdot \overline{f-x}}{\overline{a-x'} \cdot \overline{b-x'} \cdot \overline{c-x'} \cdot \overline{d-x'} \cdot \overline{e-x'} \cdot \overline{f-x'}}},$$

viz. this has the same value in all the functions.

The general property of the double ϑ -functions is that the squares of the sixteen functions are proportional to constant multiples of the sixteen functions $[a], [ab]$; but this theorem may be presented in a much more definite form, viz. we can determine, and

that very simply, the actual expressions for the constant factors; and so we can enunciate the theorem as follows; the squares of the sixteen double ϑ -functions are proportional to sixteen functions $\{a\}, \{ab\}$, where, in a notation about to be explained,

$$\{a\} = \sqrt{a} [a], \quad \{ab\} = \sqrt{ab} [ab].$$

Here in the radical $\sqrt{a}$, a is to be considered as standing in the first place for the pentad $bcdef$, which is to be interpreted as a product of differences,

$$= bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef,$$

(where bc, bd , etc., denote the differences $b - c, b - d$, etc.). Similarly, in the radical $\sqrt{ab}$, ab is to be considered as standing in the first instance for the double triad $abf \cdot cde$, which is to be interpreted as a product of differences, $= ab \cdot af \cdot bf \cdot cd \cdot ce \cdot de$, (where ab, af , etc., denote the differences $a - b, a - f$, etc.).

It is convenient to consider a, b, c, d, e, f as denoting real magnitudes taken in decreasing order: in all the products $bcdef$, etc., and in each term abf or cde of a product $abf \cdot cde$, the letters are to be written in alphabetical order; the differences bc, bd , etc., ab, af , etc., which present themselves in the several products, are thus all of them positive; and the radicals, being all of them the roots of positive quantities, may themselves be taken to be positive.

We have to consider the values of the functions $[a], [ab]$, or $\{a\}, \{ab\}$, in the case where the variables x, x' become equal to any two of the letters a, b, c, d, e, f ; it is clearly the same thing whether we have for instance $x = b, x' = c$, or $x = c, x' = b$, etc.; we have therefore to consider for x, x' the fifteen values $ab, ac, \dots, af, \dots, ef$; there is besides a sixteenth set of values x, x' each infinite, without any relation between the infinite values.

Taking this case first, x, x' each infinite, and in $[ab]$, etc., the sign $\pm$ to be $+$, we have

$$[a] = xx', \quad [ab] = \frac{4x^3 x'^3}{(x - x')^2},$$

or, attending only to the ratios of these values,

$$[a] = 1, \quad [ab] = \frac{4x^3 x'^3}{(x - x')^2},$$

where $\frac{4x^3 x'^3}{(x - x')^2}$ is infinite, and the values may finally be written

$$[a] = 0, \quad [ab] = 1;$$

whence also, for x, x' infinite,

$$\{a\} = 0, \quad \{ab\} = \sqrt{ab},$$

the radical $\sqrt{ab}$ being understood as before.

Suppose next that x, x' denote any two of the letters, for instance a, b ; then two of the functions $[a]$ vanish, viz. these are $[a], [b]$, but the remaining four functions acquire determinate values; and moreover four of the functions $[ab]$ vanish, viz. these are $[ab], [cd], [ce], [de]$, for each of which the xx' letters a, b occur in the same triad (the

double triads for the four functions are, in fact, $abf \cdot cde, cdf \cdot abe, cef \cdot abd, def \cdot abc$); but the other six functions $[ab]$, for which the letters a, b occur in separate triads, acquire determinate values.

It is important to attend to the signs: for example, if $x, x' = b, c$, we have

$$[c] = ce \cdot cb = -bc \cdot ce,$$

$$[ce] = \frac{cb \cdot eb \cdot fb}{ae \cdot be \cdot de} = \frac{cb \cdot fb}{ae \cdot de} = -\frac{bc \cdot bf}{ae \cdot de}.$$

TABLE I. OF THE VALUES OF $[a], [ab]$, ETC., FOR THE SIXTEEN SPECIAL VALUES OF x, x' .

[The original printed table is a large 16×16 array giving, for each of the sixteen special pairs $x, x' = \infty\infty, ab, ac, ad, ae, af, bc, bd, be, bf, cd, ce, cf, de, df, ef$ (across the top), the corresponding values (with proper signs) of the sixteen functions

$$[a], [b], [c], [d], [e], [f], [ab], [ac], [ad], [ae], [bc], [bd], [be], [cd], [ce], [de]$$

(down the side). Each non-zero entry is a product of two differences such as $+ab \cdot ac, -bc \cdot ef$, etc.; the diagonal entries $[a]$ for $x, x' = ab$ vanish (since one of x, x' equals a), as do entries $[ab]$ when a, b occur in the same triad of the double triad. The full table occupies one printed page and is reproduced unaltered from the original; it is used solely in deriving the corresponding table of $\{a\}, \{ab\}$ that follows.]

Here the symbols bc, ce , etc., denote differences; $[ce]$ is the product of four differences; the arrangement in two lines is for convenience only.

We thus obtain the series of values of $[a], [ab]$, etc., which although only required as subsidiary to the determination of the corresponding values of $\{a\}, \{ab\}$, I nevertheless give in a table.

The signs are given as they were actually obtained, but as we are concerned only with the ratios of the functions, it is allowable to change all the signs in any column: and it appears that there are four columns in each of which the signs are or can be made all +; whereas in each of the remaining twelve columns the signs are or can be made six of them +, the other four —.

Passing to the values of $\{a\}$, $\{ab\}$, etc., we have for example, from the ab column of the foregoing table,

$$\begin{aligned}\{c\} &= +\sqrt{c \cdot ac \cdot bc}, \\ \{d\} &= +\sqrt{d \cdot ad \cdot bd}, \\ &\vdots \\ \{ac\} &= -\sqrt{ac \cdot \frac{ac \cdot ae}{bc \cdot bf}}, \\ &\vdots\end{aligned}$$

where (since the radicals are all positive) the signs are correct: substituting for the quantities under the radical signs their full values, and squaring the rational parts in order to bring them also under the radical signs, this is

$$\begin{aligned}\{c\} &= +\sqrt{ab \cdot ad \cdot ae \cdot af \cdot bd \cdot be \cdot bf \cdot de \cdot df \cdot ef \cdot ac^2 \cdot bc^2}, \\ \{d\} &= +\sqrt{ab \cdot ac \cdot ae \cdot af \cdot bc \cdot be \cdot bf \cdot ce \cdot cf \cdot ef \cdot ad^2 \cdot bd^2}, \\ \{ac\} &= -\sqrt{ac \cdot af \cdot cf \cdot bd \cdot be \cdot de \cdot ac^2 \cdot ae^2 \cdot bc^2 \cdot bf^2},\end{aligned}$$

where all the expressions of this (the ab -column) have a common factor,

$$ac \cdot ad \cdot ae \cdot af \cdot bc \cdot bd \cdot be \cdot bf.$$

Omitting this factor, we find

$$\begin{aligned}\{c\} &= +\sqrt{ab \cdot ac \cdot bc \cdot de \cdot df \cdot ef}, \\ \{d\} &= +\sqrt{ab \cdot ad \cdot bd \cdot ce \cdot cf \cdot ef}, \\ \{ac\} &= -\sqrt{ad \cdot ae \cdot de \cdot bc \cdot bf \cdot cf};\end{aligned}$$

viz. recurring to the foregoing condensed notation, this is

$$\begin{aligned}\{c\} &= +\sqrt{de}, \\ \{d\} &= +\sqrt{ce}, \\ \{ac\} &= -\sqrt{bc},\end{aligned}$$

and, in fact, the terms in the several columns have only the ten values $\sqrt{ab}$, $\sqrt{ac}$, etc. each with its proper sign. I repeat the meaning of the notation: ab stands in the first instance for the double triad $abf \cdot cde$, and then this denotes a product of differences $ab \cdot af \cdot bf \cdot cd \cdot ce \cdot de$. We have thus the following table in which I have in several cases changed the signs of entire columns.

TABLE II. OF THE FUNCTIONS $\{a\}, \{ab\}$, ETC., FOR THE SIXTEEN SPECIAL VALUES OF x, x' .

[The original printed Table II. is a 16×16 array showing, for each of the sixteen special pairs $x, x' = \infty\infty, ab, ac, ad, ae, af, bc, bd, be, bf, cd, ce, cf, de, df, ef$ (across the top), the corresponding values of the sixteen functions

$$\{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{f\}, \{ab\}, \{ac\}, \{ad\}, \{ae\}, \{bc\}, \{bd\}, \{be\}, \{cd\}, \{ce\}, \{de\}$$

(down the left side). Each non-zero entry is one of the ten radicals $\pm\sqrt{ab}, \pm\sqrt{ac}, \dots, \pm\sqrt{de}$ with its proper sign; in many entries the sign of the entire column has been reversed (as compared with the values directly derived from Table I) so as to present the functions in their most convenient form. The full table occupies one printed page; it shows that the sixteen functions $\{a\}, \{ab\}$, take, over the sixteen special pairs x, x' , only the ten values $\sqrt{ab}, \sqrt{ac}, \dots, \sqrt{de}$ each with a proper sign.]

Referring now to Göpel's memoir, *Crelle*, t. XXXV. (1847), pp. 277–312, we have the sixteen double ϑ -functions

$$P, P_1, P_2, P_3; \ iQ, Q_1, iQ_2, Q_3; \ iR, iR_1, R_2, R_3; \ S, iS_1, iS_2, S_3,$$

where the six functions affected with the $i (= \sqrt{-1})$ are odd functions, vanishing for the values $u = 0, u' = 0$ of the arguments. It is convenient to take $x = \infty, x' = \infty$ as the values of x, x' corresponding to these values $u = 0, u' = 0$: the expressions $\{a\}$ will thus correspond to the six squares $-Q^2, -Q_2^2, -R^2, -R_1^2, -S_1^2, -S_2^2$, and the expressions $\{ab\}$ to the remaining ten squares $P^2, P_1^2, \dots, S_3^2$; and after some *tâtonnement*, I succeed in establishing the correspondence as follows:

$$\begin{aligned} & -S_2^2, -S_1^2, -R_1^2, -R^2, -Q_2^2, -Q^2, P_3^2, P_1^2, P^2, -S_3^2, P_2^2, P_1^2, S_1^2, Q_1^2, R_2^2, R_3^2, \\ & = \{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{f\}, \{ab\}, \{ac\}, \{ad\}, \{ae\}, \{bc\}, \{bd\}, \{be\}, \{cd\}, \{ce\}, \{de\}, \end{aligned}$$

viz. the sixteen squared double ϑ -functions are proportional to the sixteen expressions $\{a\}, \{ab\}$, as hereby appearing.

We have, after Göpel (l.c. p. 283), a table showing how the ratios of the double ϑ -functions are altered, when the arguments are increased by the quarter-periods

$$A, B, A+B, K, K+A, K+B,$$

that is, when u, u' are simultaneously changed into $u+A, u'+A'$, or into $u+B, u'+B'$, etc. If instead, we consider the squared functions, the table is very much simplified, inasmuch as in place of the coefficients $\pm 1, \pm i$, it will contain only the coefficients ± 1 : and we may complete the table by extending it to all the combinations $0, A, B, A+B, K, K+A, K+B, K+A+B, L, L+A, L+B, L+A+B, K+L, K+L+A, K+L+B, K+L+A+B$ of the quarter-periods: we have thus a table included in the annexed Table III., viz. attending herein only to the capital letters P, Q, R, S , the sixteen columns of the table show how the ratios of the terms $-S_2^2, -S_1^2$, etc., of the first column are altered when the arguments are increased by the foregoing combinations of quarter-periods, as indicated by the headings $0, A, B$, etc., of the several columns.

TABLE III. OF THE SIXTEEN FORMS OF THE SQUARED DOUBLE ϑ -FUNCTIONS.

[Table III. is a 16×16 array whose left-hand column lists the sixteen squared functions

$$-S_2^2, -S_1^2, -R_1^2, -R^2, -Q_2^2, -Q^2, P_3^2, P_1^2, P^2, -S_3^2, P_2^2, P_1^2, S_1^2, Q_1^2, R_2^2, R_3^2,$$

and whose other fifteen columns are headed

$$A, B, A+B, K, K+A, K+B, K+A+B, L, L+A, L+B, L+A+B, K+L, K+L+A, K+L+B, K+L+A+B,$$

Each entry shows how the squared function in the row is transformed under the increment of the arguments by the quarter-period combination labelling the column, and is given as $\pm$ (another squared function) followed by its value as one of the symbols $a, b, c, \dots, ab, ac, \dots, de$ from the foregoing scheme. At the foot of each column is inserted the suffix (∞, ab, cd , etc.) referring to the columns of Table II. The full 16×16 table is reproduced unaltered from the original; the entries are used in the verification on the next page.]

But I have also in the table inserted the values to which $-S_2^2, -S_1^2$, etc., are respectively proportional, viz. the table runs $-S_2^2 = a, -S_1^2 = b$, etc., (read $-S_2^2 = \{a\}, -S_1^2 = \{b\}$, etc., the brackets $\{ \}$ having been for greater brevity omitted throughout the table), and where it is of course to be understood that $-S_2^2, -S_1^2$, etc., are proportional only, not absolutely equal to $\{a\}, \{b\}$, etc. And I have also at the foot of the several columns inserted suffixes $\infty\infty, ab, cd$, etc., which refer to the columns of Table II.

Comparing the first with any other column of the table, for instance with the second column, the two columns respectively signify that

$$\begin{aligned} -S_2^2(u) &= \{a\}, & -S_2^2(u+A) &= -\{ae\}, \\ -S_1^2(u) &= \{b\}, & -S_1^2(u+A) &= -\{be\}, \\ &\vdots & &\vdots \\ Q_1^2(u) &= \{ab\}, & Q_1^2(u+A) &= -\{e\}, \end{aligned}$$

where, as before, the sign $=$ means only that the terms are proportional; u is written for shortness instead of (u, u') , and so $u+A$ for $(u+A, u'+A')$, etc.: the variables in the functions $\{a\}, \{b\}$, etc. are in each case x, x' . But if in the second column we write $u-A$ for A , then the variables x, x' will be changed into new variables y, y' , or the meaning will be

$$\begin{aligned} x, x' & & y, y' \\ -S_2^2(u) &= \{a\}, & -S_2^2(u) &= -\{ae\}, \\ -S_1^2(u) &= \{b\}, & -S_1^2(u) &= -\{be\}, \\ &\vdots & &\vdots \\ Q_1^2(u) &= \{ab\}, & Q_1^2(u) &= -\{e\}, \end{aligned}$$

so that, omitting from the table the terms which contain the capital letters P, Q, R, S , except only the outside left-hand column $-S_2^2, -S_1^2$, etc., the table indicates that these functions $-S_2^2, -S_1^2$, etc., are proportional to the functions $\{a\}, \{b\}$, etc., of x, x' given in the first column; also to the functions $-\{ae\}, -\{be\}$, etc., of y, y' given in the second column; also to the functions $-\{ae\}, -\{be\}$, etc., of z, z' given in the third column; and so on, with a different pair of variables in each of the 16 columns.

Thus comparing any two columns, for instance the first and second, it appears that we can have simultaneously

$$\begin{aligned} x, x' & & y, y' \\ \{a\} &= -\{ae\}, \\ \{b\} &= -\{be\}, \\ &\vdots \\ \{ab\} &= -\{e\}, \end{aligned}$$

(fifteen equations, since the meaning is that the terms are only proportional, not absolutely equal), equivalent to two equations serving to determine x and x' in terms of y and y'

or conversely y and y' in terms of x and x' . The functions in each column form in fact 16 sixes, such that any four belonging to the same six are linearly connected; and in any such linear relation between four functions in the left-hand column, substituting for these their values as functions in the right-hand column, we have the corresponding relations between four functions out of a set of six belonging to the right-hand column, or we have an identity $0 = 0$. I will presently verify this in a particular case.

If in any column we give to the variables the values x, x' , we obtain for the terms in the column the values which the terms of the first column assume on giving to x, x' the values shown at the foot of the column in question; thus, in the second column giving to the variables the values c, d , the column becomes

$$-\sqrt{be}, -\sqrt{ae}, 0, 0, -\sqrt{ab}, \sqrt{cd}, 0, \sqrt{ad}, \sqrt{ac}, 0, \sqrt{bd}, \sqrt{bc}, 0, 0, \sqrt{de}, \sqrt{ce}$$

which is, in fact, the cd -column of Table II.: this is of course as it should be, for the values in question are those of the functions $-S_2^2, -S_1^2$, etc., on writing therein

$$x, x' = c, d.$$

The formulae show that

$$\sqrt{ab}, \sqrt{ac}, \sqrt{ad}, \sqrt{ae}, \sqrt{bc}, \sqrt{bd}, \sqrt{be}, \sqrt{cd}, \sqrt{ce}, \sqrt{de},$$

are, in fact, proportional to

$$p_2^2, \varpi_2^2, \varpi_1^2, \varpi_3^2, \sigma_3^2, p_1^2, p_3^2,$$

($k_1, k_2, \dots$ are Göpel's $k', k'', \dots$). This gives rise to a remarkable theorem, for the ten squares are functions of only four quantities $\alpha, \beta, \gamma, \delta$ (Göpel's t, u, v, w). For greater clearness, I introduce single letters $A, B, \dots, J$, and write

$$\begin{aligned} A = abc \cdot def &= (\sqrt{def})^2 = p_2^2, &= \frac{1}{4}(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)^2, \\ B = abd \cdot cef &= (\sqrt{cef})^2 = p_2'^2, &= \frac{1}{4}(\alpha\gamma + \beta\delta)^2, \\ C = abe \cdot cdf &= (\sqrt{cdf})^2 = k_3, &= \frac{1}{4}(\alpha\delta + \beta\gamma)^2, \\ D = abf \cdot cde &= (\sqrt{cde})^2 = k_1, &= \frac{1}{4}(\alpha^2 - \beta^2 - \gamma^2 + \delta^2)^2, \\ E = acd \cdot bef &= (\sqrt{bef})^2 = \varpi_3^2, &= \frac{1}{4}(\alpha\beta + \gamma\delta)^2, \\ F = ace \cdot bdf &= (\sqrt{bdf})^2 = \varpi_2^2, &= \frac{1}{4}(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)^2, \\ G = acf \cdot bde &= (\sqrt{acf})^2 = \varpi_1^2, &= \frac{1}{4}(\alpha\delta - \beta\gamma)^2, \\ H = ade \cdot bcf &= (\sqrt{bcf})^2 = \varpi_1'^2, &= \frac{1}{4}(\alpha\gamma - \beta\delta)^2, \\ I = adf \cdot bce &= (\sqrt{adf})^2 = \sigma_3^2, &= -\frac{1}{4}(\alpha\beta - \gamma\delta)^2, \\ J = aef \cdot bdc &= (\sqrt{aef})^2 = \sigma_3'^2, &= \frac{1}{4}(\alpha^2 + \beta^2 - \gamma^2 - \delta^2)^2; \end{aligned}$$

viz. it has to be shown that $A, B, \dots, J$, considered as given functions of the six letters a, b, c, d, e, f , are really functions of four quantities $\alpha, \beta, \gamma, \delta$; or, what is the same thing, that $A, B, \dots, J$, considered as functions of a, b, c, d, e, f , satisfy all those relations which they satisfy when considered as given functions of $\alpha, \beta, \gamma, \delta$.

Now considering them as given functions of $\alpha, \beta, \gamma, \delta$, they ought to satisfy six relations; and inasmuch as, so considered, they are, in fact, linear functions of

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4, \quad \alpha^2\beta^2 + \gamma^2\delta^2, \quad \alpha^2\gamma^2 + \beta^2\delta^2, \quad \alpha^2\delta^2 + \beta^2\gamma^2, \quad \alpha\beta\gamma\delta,$$

five of these relations will be linear: there is a sixth non-linear relation, expressible in a variety of different forms, one of them, as is easily verified, being

$$\sqrt{AJ} \pm \sqrt{CG} \pm \sqrt{DF} = 0.$$

Now considering $A, B, \dots, J$ as given functions of a, b, c, d, e, f , there exist between them linear relations which may be obtained by the consideration of identities of the form

$$\overline{abcd} = 0, \quad \overline{abcdef} = 0,$$

where the left-hand side is used for shortness to denote the determinant

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ a, & b, & c, & d \\ a^2, & b^2, & c^2, & d^2 \end{vmatrix} = 0, \quad \begin{vmatrix} 1, & 1, & 1, & 1, & 1, & 1 \\ f, & a, & b, & c, & d, & e \\ f^2, & a^2, & b^2, & c^2, & d^2, & e^2 \end{vmatrix} = 0.$$

We thus obtain between them a system of fifteen linear relations, which present themselves in the form

- (1) $A - J + E - B = 0,$
- (2) $-A - I + F - C = 0,$
- (3) $A - H + G - D = 0,$
- (4) $-B - G + H + C = 0,$
- (5) $B - F + I + D = 0,$
- (6) $C - E + J - D = 0,$
- (7) $-E - D - H + F = 0,$
- (8) $E - C + G - I = 0,$
- (9) $F - B - J - G = 0,$
- (10) $H - A + J - I = 0,$
- (11) $-J + D - G + I = 0,$
- (12) $J + C - F + H = 0,$
- (13) $I + B - E - H = 0,$
- (14) $G + A + E - F = 0,$
- (15) $D - A + B - C = 0,$

and these are all included in the equations (10), (4), (12), (15), (6), which serve to express G, B, E, F, I in terms of D, H, C, A, J , i.e. ac, ce, eb, bd, da in terms of $G = ab, bc, cd, de, ea$, if for the moment we write $G = ac$, etc. But the five linear relations in question are, it is at once seen, satisfied by $A, B, \dots, J$ considered as given functions of $\alpha, \beta, \gamma, \delta$.

The equation $\sqrt{AJ} \pm \sqrt{CG} \pm \sqrt{DF} = 0$, substituting for $A, B, \dots$, their values in terms of a, b, c, d, e, f , becomes

$$\sqrt{abc \cdot def \cdot aef \cdot bcd} \pm \sqrt{abf \cdot cde \cdot ace \cdot bdf} \pm \sqrt{abe \cdot cdf \cdot acf \cdot bde} = 0,$$

which (omitting common factors) becomes $\pm \sqrt{bc^2 \cdot ef^2} \pm \sqrt{bf^2 \cdot ce^2} \pm \sqrt{be^2 \cdot cf^2} = 0$; or, taking the proper signs, this is the identity $bc \cdot ef + be \cdot cf - bf \cdot ce = 0$.

It is to be noticed that

$$\begin{array}{lll} \delta^2 + \alpha^2 - \beta^2 - \gamma^2, & 2(\alpha\beta - \gamma\delta), & 2(\gamma\alpha + \beta\delta), \\ 2(\alpha\beta + \gamma\delta), & \delta^2 + \beta^2 - \gamma^2 - \alpha^2, & 2(\beta\gamma - \alpha\delta), \\ 2(\gamma\alpha - \beta\delta), & 2(\beta\gamma + \alpha\delta), & \delta^2 + \gamma^2 - \alpha^2 - \beta^2, \end{array}$$

each divided by $\delta^2 + \alpha^2 + \beta^2 + \gamma^2$, form a system of coefficients in the transformation between two sets of rectangular coördinates. We have therefore

$$\begin{array}{l} \sqrt{ab}, \sqrt{ad}, \sqrt{ce}, \\ \sqrt{be}, \sqrt{de}, \sqrt{ac}, \\ \sqrt{bf}, \sqrt{cd}, \sqrt{ae}, \end{array}$$

each divided by $\sqrt{bd}$ and the several terms taken with proper signs, as a system of coefficients in the transformation between two sets of rectangular axes: a result which seems to be the same as that obtained by Hesse in the Memoir, "Transformations-Formeln für rechtwinklige Raum-Coordinaten"; *Crelle*, t. LXIII. (1864), pp. 247–251.

The composition of the last mentioned system of functions is better seen by writing them under the fuller form $\sqrt{abf \cdot cde}$, etc.; viz. omitting the radical signs, the terms are

$$\begin{array}{lll} abf \cdot cde, & adf \cdot bce, & abd \cdot cef, \\ bef \cdot acd, & def \cdot abc, & acf \cdot bde, \\ bcf \cdot ade, & cdf \cdot abe, & aef \cdot bcd, \end{array}$$

each divided by $bdf \cdot ace$; or, in an easily understood algorithm, the terms are

$$\begin{array}{lll} a \cdot ce & bf \cdot d & df \cdot b \quad bd \cdot f \\ e \cdot ac & bf \cdot d & df \cdot b \quad bd \cdot f \\ c \cdot ae & bf \cdot d & df \cdot b \quad bd \cdot f \end{array}$$

each divided by $bdf \cdot ace$.

Reverting to the before-mentioned comparison of the first and second columns of Table III., four of the equations are

$$\begin{array}{ll} x, x' & y, y' \\ \{c\} = \{d\}, & \text{that is, } \sqrt{c}[c] = \sqrt{d}[d], \\ \{d\} = \{c\}, & \text{that is, } \sqrt{d}[d] = \sqrt{c}[c], \\ \{e\} = -\{ab\}, & \text{that is, } \sqrt{e}[e] = -\sqrt{ab}[ab], \\ \{f\} = -\{cd\}, & \text{that is, } \sqrt{f}[f] = -\sqrt{cd}[cd]; \end{array}$$

viz. the four terms on the left-hand side are not absolutely equal, but only proportional, to those on the right-hand side. Substituting for $\sqrt{c}$, $\sqrt{d}$, etc., their values, and introducing on the right-hand side the factor

$$\sqrt{ac \cdot bc \cdot ce \cdot cf \cdot ad \cdot bd \cdot de \cdot df},$$

the equations become

$$\begin{array}{ll} xx' & yy' \\ [c] = ac \cdot bc \cdot ce \cdot ef [d], & \\ [d] = ad \cdot bd \cdot de \cdot df [c], & \\ [e] = -ce \cdot de [ab], & \\ [f] = -cf \cdot df [cd]. & \end{array}$$

The functions on the left-hand satisfy the identity

$$ef[c] - ef[d] + fcd[e] - cde[f] = 0,$$

or, as this may also be written,

$$def[c] - cef[d] + cdf[e] - cde[f] = 0.$$

Hence substituting the right-hand values, the whole equation divides by $ce \cdot de \cdot cf \cdot df$; omitting this factor, it becomes

$$ef \cdot ac \cdot bc [d] - ef \cdot ad \cdot bd [c] - cd \{[ab] - [cd]\} = 0,$$

where the variables are y, y' : it is to be shown that this is in fact an identity, and (as it is thus immaterial what the variables are) I change them into x, x' .

We have $ac \cdot bc [d] - ad \cdot bd [c]$

$$= (a - c)(b - c)(d - x)(d - x') - (a - d)(b - d)(c - x)(c - x')$$

suppose,

$$\begin{aligned} &= (c - d) \begin{vmatrix} 1, & x + x', & xx' \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix} \\ &= cd [xx'abcd]. \end{aligned}$$

We have moreover

$$\begin{aligned} [ab] - [cd] &= ef \frac{1}{(x - x')^2} \{abf' \cdot c'd'e' + a'b'f \cdot cde - cdf' \cdot a'b'e' - c'd'f \cdot abe\} \\ &= \frac{1}{(x - x')^2} (abc'd' - a'b'cd)(ef' - e'f), \end{aligned}$$

where for the moment a, b, a' , etc., are written to denote $a - x, b - x, a - x'$, etc.; we have then

$$e'f - ef' = (e - x')(f - x) - (e - x)(f - x') = -(e - f)(x - x') = -ef(x - x'),$$

and

$$\begin{aligned} abc'd' - a'b'cd &= (a - x)(b - x)(c - x')(d - x') - (a - x')(b - x')(c - x)(d - x) \\ &= (x - x') \begin{vmatrix} 1, & x + x', & xx' \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix} = -(x - x')[xx'abcd]. \end{aligned}$$

Hence $[ab] - [cd] = ef[xx'abcd]$, and the equation to be verified becomes

$$(ef \cdot cd - cd \cdot ef)[xx'abcd] = 0,$$

viz. this is, in fact, an identity.

Cambridge, 14 March, 1877.

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ON THE 16-NODAL QUARTIC SURFACE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXIV. (1877), pp. 238–241.]

Prof. Borchardt in the Memoir “Ueber die Darstellung u. s. w.” Crelle, t. LXXXIII. (1877), pp. 234–243, shows that the coordinates x, y, z, w may be taken as proportional to four of the double ϑ -functions, and that the equation of the surface is then Göpel’s relation of the fourth order between these four functions: and he remarks at the end of the memoir that it thus appears that the coordinates x, y, z, w of a point on the surface can be expressed as proportional to algebraic functions, involving square roots only, of two arbitrary parameters ξ, ξ' .

It is interesting to develop the theory from this point of view. Writing, as in my paper, “Further investigations on the double ϑ -functions,” pp. 220–233, [663],

$$[a] = aa', \quad [b] = bb', \quad [c] = cc', \quad [d] = dd', \quad [e] = ee', \quad [f] = ff',$$

$$[ab] = \frac{1}{(\xi - \xi')^2} (\sqrt{abf \cdot c'd'e'} - \sqrt{a'b'f' \cdot cde})^2,$$

etc.,

where on the right-hand sides $a, b, \dots, a', \dots$ denote $a - \xi, b - \xi, \dots, a - \xi', \dots$ (ξ, ξ' being here written in place of the x, x' of my paper), then the sixteen functions